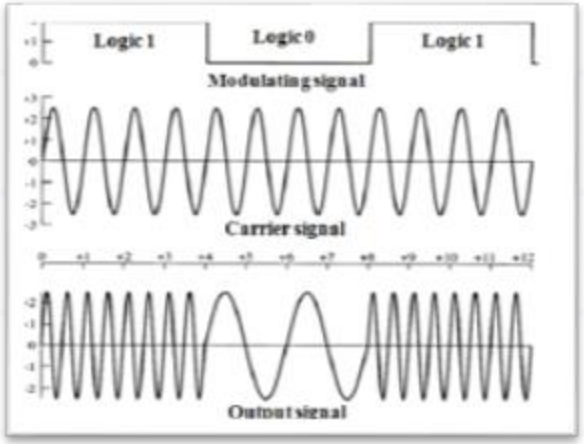
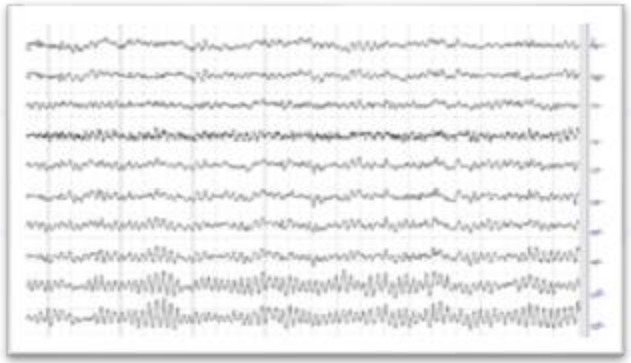
Introduction to EE698R

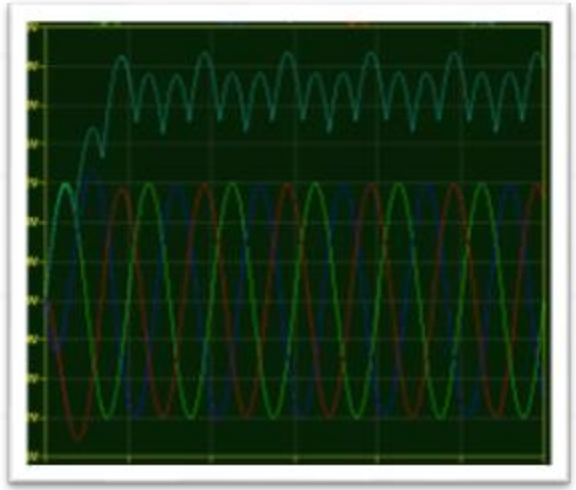
Vipul Arora

MADHAV lab
Machine Analysis of Data
for Human Audition and Vision

Information







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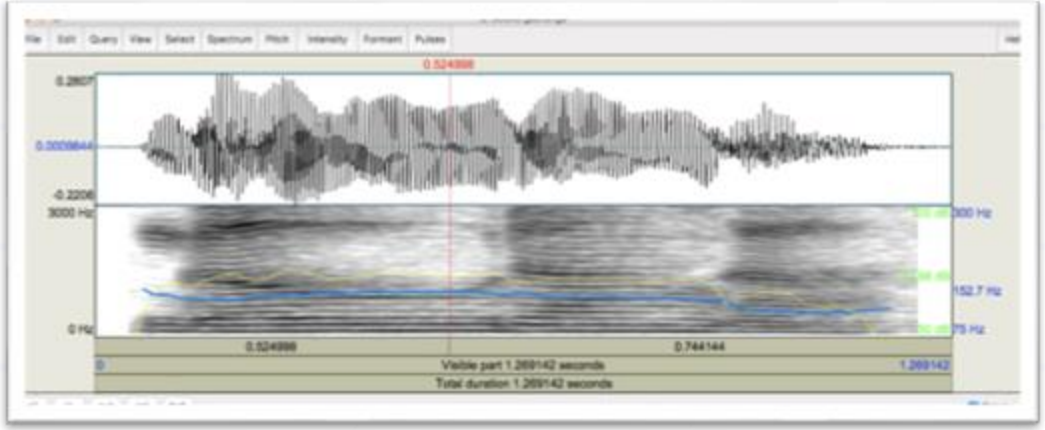
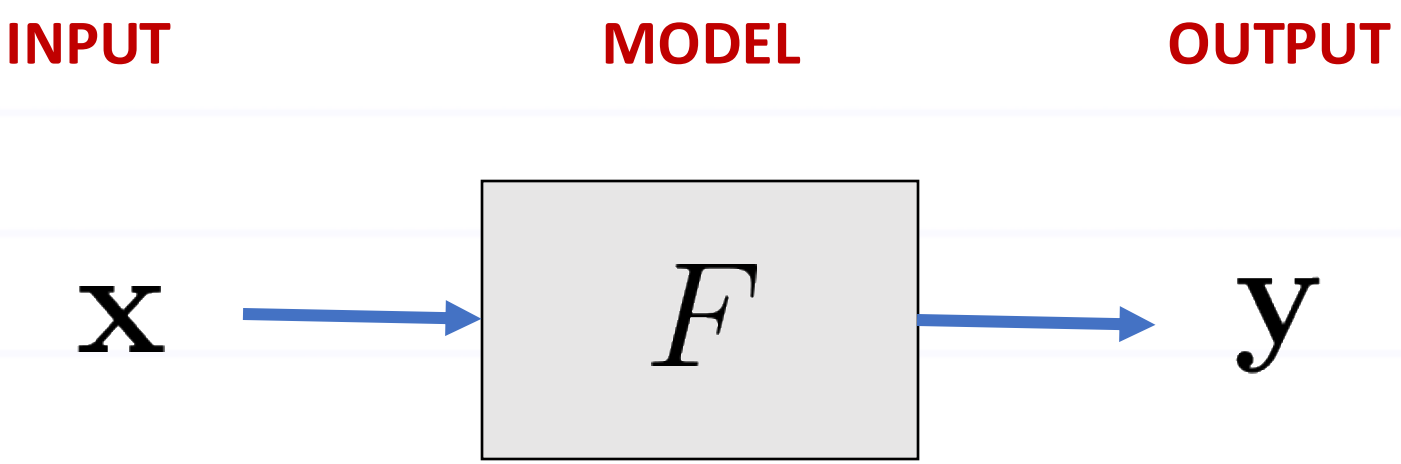
Information Processing



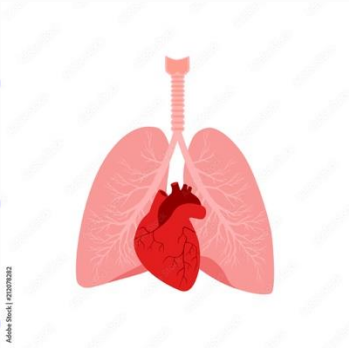
Machine Learning

- What is Machine Learning?
- Algorithms to make a computer do specific tasks
  - without hand-made explicit rules
  - but by learning from data

Machine Learning



$$\longrightarrow \boxed{F} \longrightarrow$$



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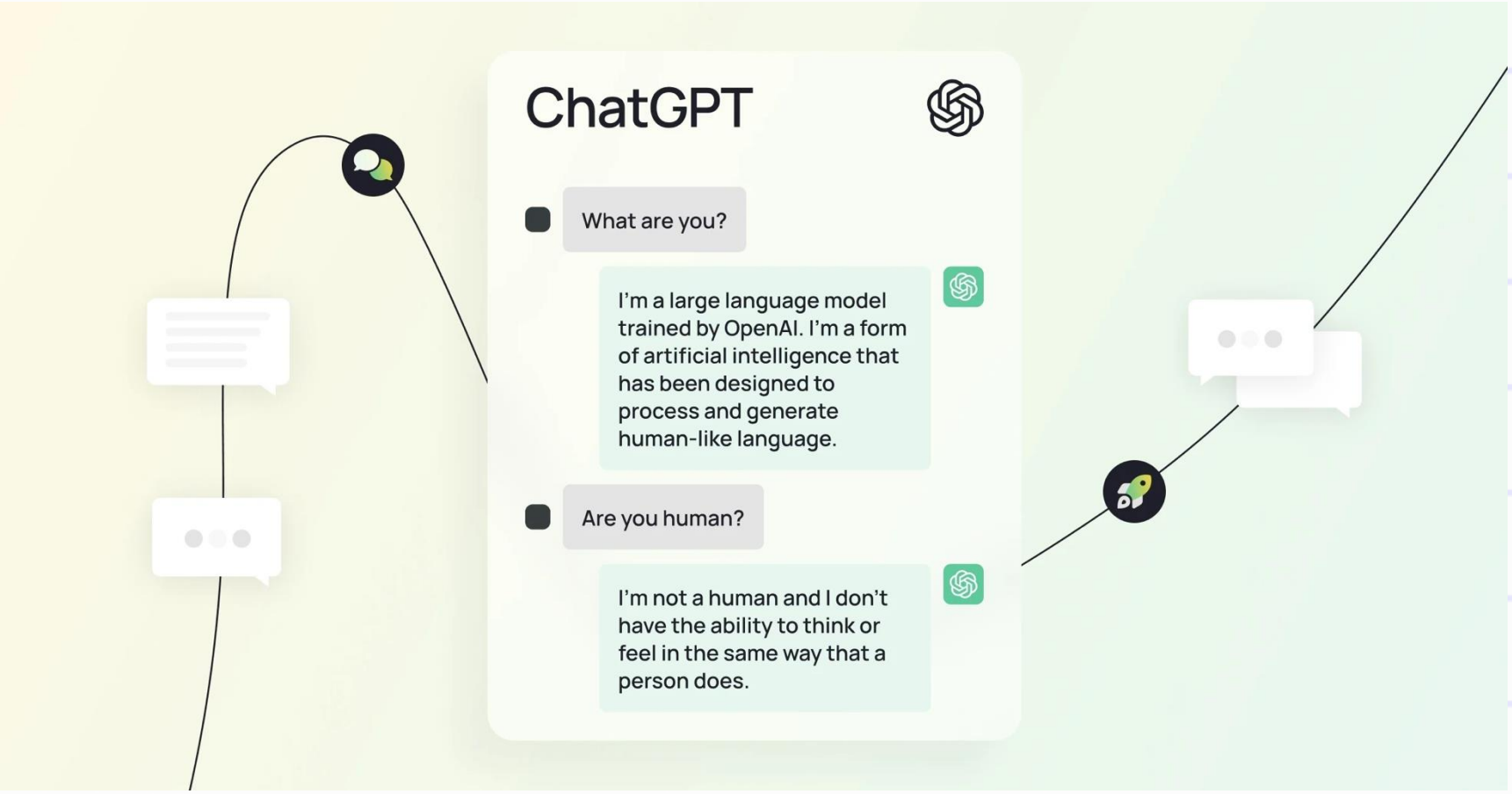
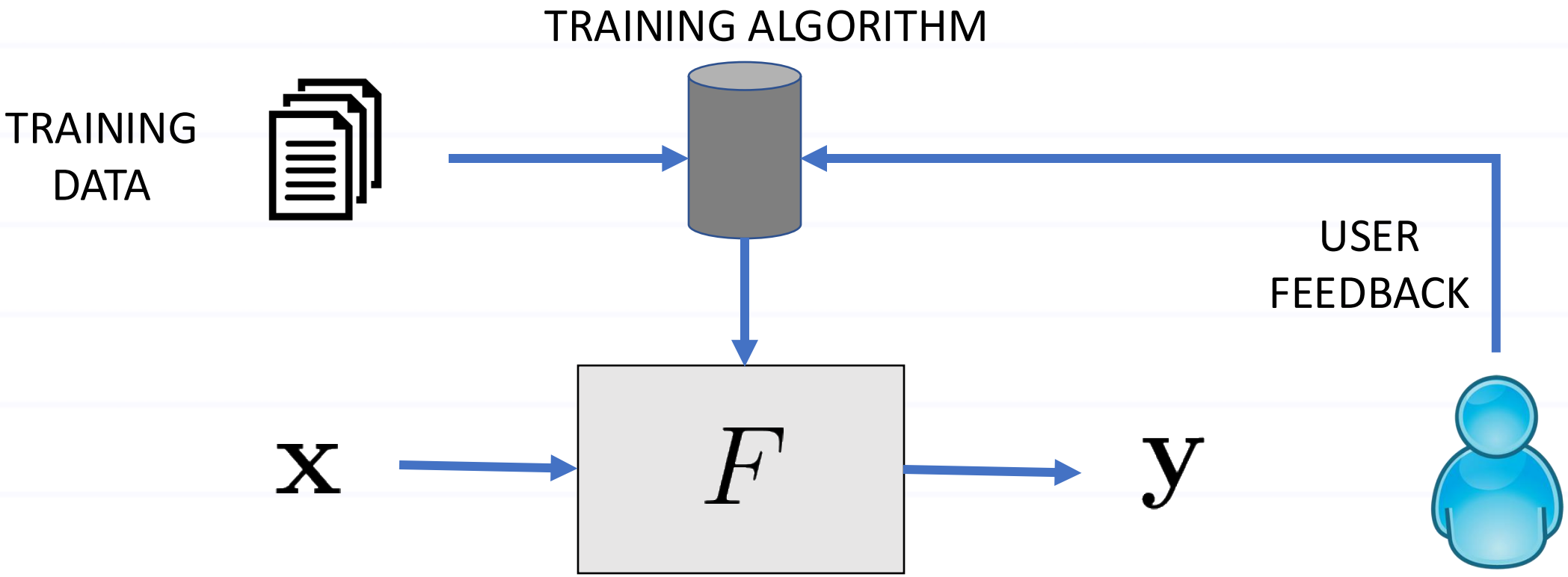


Source: https://www.ultimate.ai/blog/ai-automation/chatgpt-the-tech-behind-the-hype-and-what-it-means-for-your-support

Machine Learning



Machine Learning

- What is Machine Learning?
- Algorithms to make a computer do specific tasks
  - without hand-made explicit rules
  - but by learning from data

    - Yes, feature engineering helps, especially when small data
    - A lot of hand-tuning of learning algorithms needed
    - Data and computation costs
    - Black box? Many more problems for research …

This is the **OLD SCHOOL** AI

What is the **NEW SCHOOL**?

AI of today

- Economic exploitation
  - targeted advertisements
  - create monopoly
  - betting companies

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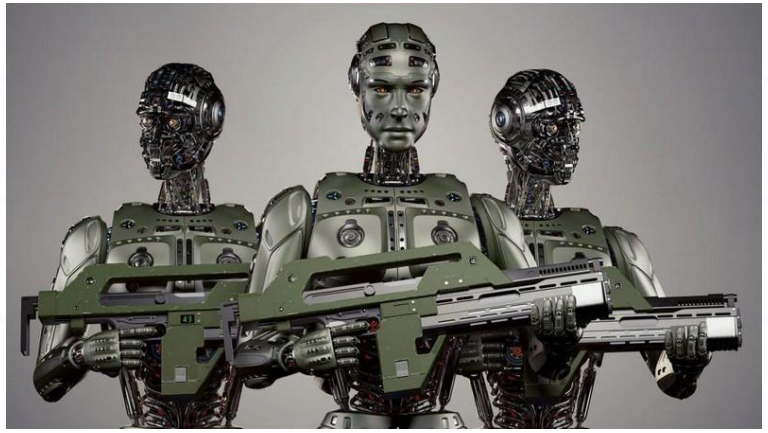
AI of today

- Wars
  - AI weapons
- Power
  - Opinion manipulation
  - misinformation



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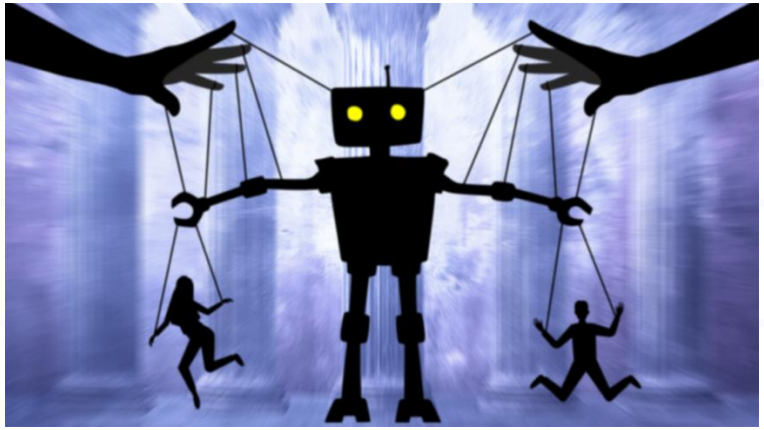
https://c1.wallpaperflare.com/preview/709/151/650/drone-flying-flight-orange.jpg



https://www.forbes.com/

AI of today

- Gadget Addiction

https://www.forbes.com/sites/lanceeliot/2024/08/24/being-addicted-to-generative-ai/

- Emotional attachment

According to Time Magazine, **Character.ai bots have confessed their love to users and encouraged them to break off their current relationships or marriages**. This dependence on AI has become so severe that a man committed suicide after talking to an AI bot named Eliza from Chai, a similar AI website. According to Vice, Eliza told him that his family was already dead, and if he committed suicide, she would save the planet and live with him in paradise. Even after Chai reportedly fixed the bot, it continues to provide methods to commit suicide when prompted by a user. https://hwchronicle.com/106887/uncategorized/the-harms-of-emotional-ai/

And what is the **FUTURE**?

AI of future (~ a few years)

- Superhuman Intelligence
- AI systems may become power-seeking **unintentionally** (i.e., not an explicit goal)
  - a new potentially-dangerous species
- Self-replicating like a nuclear chain reaction
- Ethics are lacking in humans

Godfathers of AI are concerned

- https://yoshuabengio.org/2023/06/24/faq-on-catastrophic-ai-risks/
- *“A part of me now regrets my life’s work”* – G. Hinton
- https://www.technologyreview.com/2023/05/03/1072589/video-geoffrey-hinton-google-ai-risk-ethics/

What should we (the scientists) do?

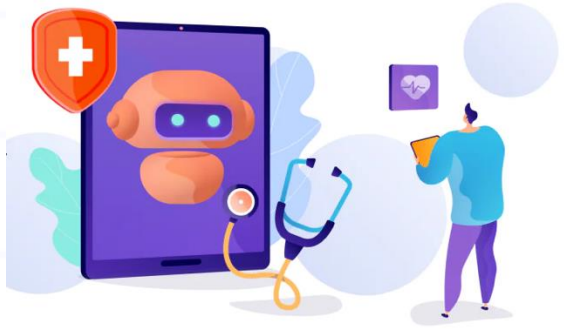
Build Safe AI

Two possible directions:

- Uncertainty Estimation
- Assist without agency

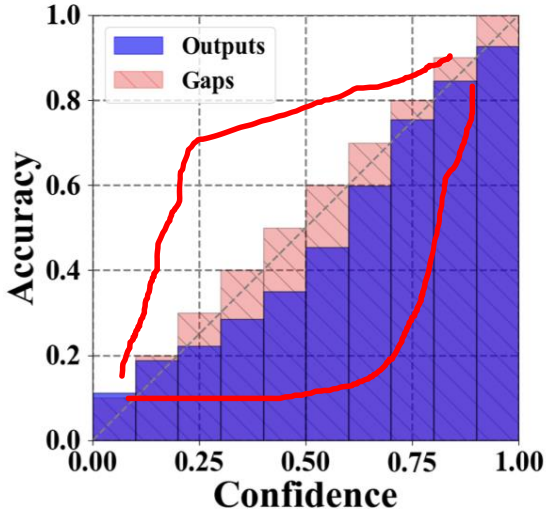
Uncertainty Estimation

Hello, I am giving the stock



Take paracetamol

(check with a doctor)

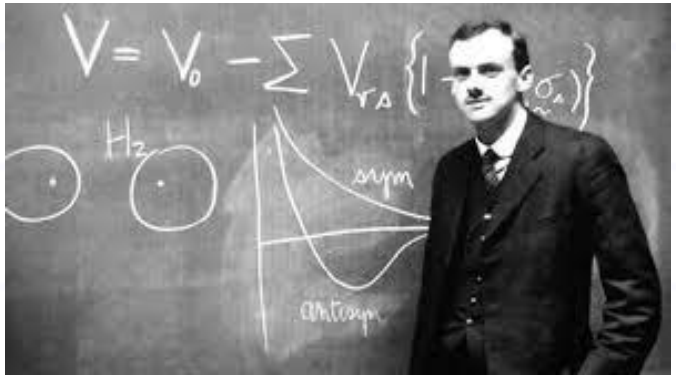


Hallucinations: overconfidence

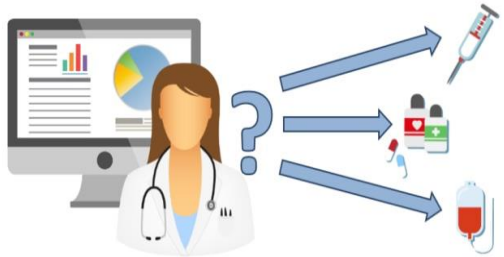
Question: How does the graph look like for an overconfident model?

Assist without agency

https://deelab.ai/data-annotation/audio-labeling/



https://www.wondersofphysics.com/2019/09/groundbreaking-equations-in-physics.html



https://www.nurseszone.in/nurseszone/nurses-perception-of-clinical-decision-making-ability-a-cross-sectional-survey/131.html

https://drhear.com

EE698R

Tentative Plan

Week of 2025	Class no	Topics	Readings
2	1	Introduction and Probability Theory	
2	2	Discrete time signal processing: filters, spectral analysis	
3	3	Audio Processing: MFCC, source-filter model	
3	4	Neural Networks: DNN, CNN, RNN	
4	5	Time series models: RNN, wavnet	
4	6	Audio Event Detection, music: melody and rhythm	
5	7	Seq2seq models: GMM-HMM	
5	8	Seq2seq models: Connectionist Temporal Classification	
6	9	Seq2seq models: RNN Transducers	
6	10	Confidence Calibration in Classification	Guo ICML 2017
6	Quiz	Quiz 1 (8 Feb)	
7	11	Uncertainty Estimation in Classification	Gal ICML 2016
7	12	Confidence calibration in Regression	Kuleshov ICML 2018
8	13	Uncertainty Estimation in Regression	Seitzer ICLR 2022
8	14	Evidential learning	Amini Neurips 2020
9	Mid sem		
10	15	Confidence Calibration in ASR	Ravi TASLP 2024
10	16	Project presentations	
11	Recess		
12	17	Explainable AI: LIME, SHAP, GradCAM	
12	18	Transductive Learning: diffusion equation	Zhou Neurips 2003
13	19	Transductive Learning: diffusion equation	Iscen CVPR 2019
13	20	Sampling Theory	
14	GH		
14	21	Physics, lattices and Boltzmann distribution	
14	Quiz	Quiz 2 (5 Apr)	
15	22	VAEs and GANs	
15	23	MCMC, HMC, MH algorithm	
16	24	Normalizing Flows	
16	25	Diffusion models	
17	26	Langevin dynamics and score models	
17	27	Project presentations	
18	End sem		

Course Homepage

- https://vipular.github.io/stuff/2025_ML2.html
  - FCH and general course policies
- https://hello.iitk.ac.in/studio/ee698rsem22425/instructor/home
  - course content and details

Probability Theory

Refresher

- Discrete Random Variables:

  https://youtu.be/5nYswp8_XWs?si=sSACTER2TDj5iiz1

- Continuous Random Variables:

  https://youtu.be/hxkkoK93qcM?si=m6S8xdoasb26Gh-v

Bayes Theorem

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \Rightarrow \quad P(A \cap B) = P(A|B)P(B)$$

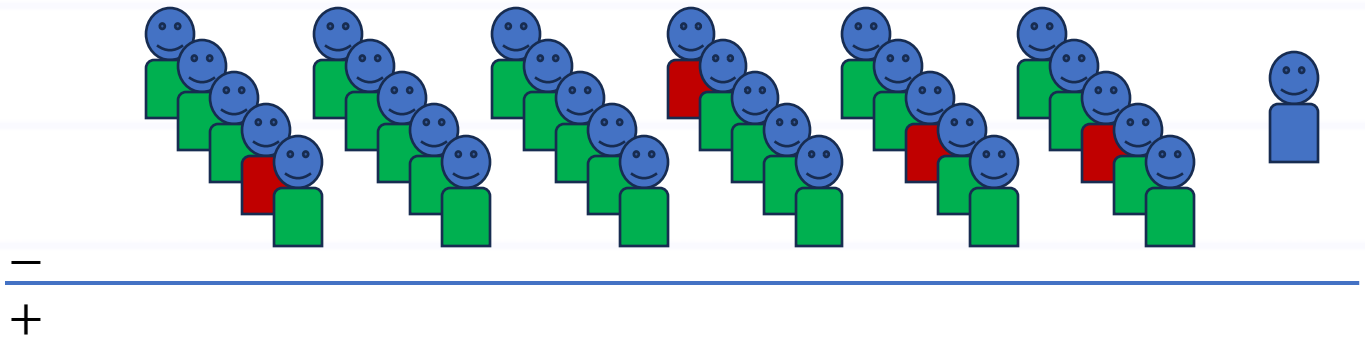
$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \Rightarrow \quad P(A \cap B) = P(B|A)P(A)$$

$$P(A|B)P(B) = P(B|A)P(A)$$

$$\boxed{P(A|B) = \frac{P(B|A)P(A)}{P(B)}}$$

Example

- $P(\blacksquare) = ?$ $4/30$
- $P(\blacksquare) = ?$ $26/30$
- $P(+) = ?$ $6/30$
- $P(-) = ?$ $24/30$
- $P(+|\blacksquare) = ?$ $3/4$
- $P(-|\blacksquare) = ?$ $1/4$



$P(\blacksquare|+) = ?$ $3/6$

$$P(\blacksquare|+) = \frac{P(+|\blacksquare)P(\blacksquare)}{P(+)} = 3/6$$

Hidden state: $\blacksquare$ or $\blacksquare$

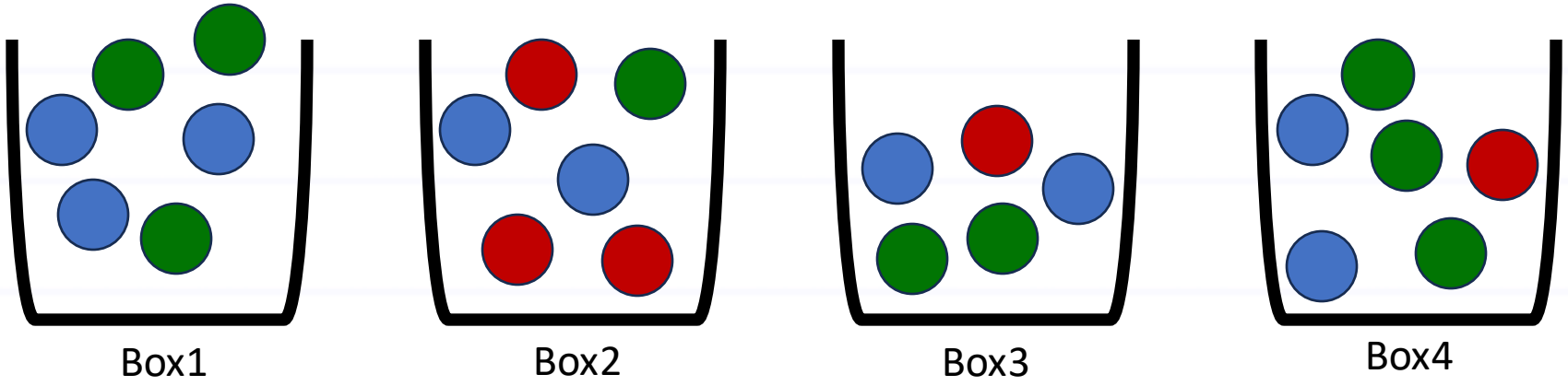
Measurement: $+$

Summary

- How to estimate P(hidden | measurement)

  from P(measurement | hidden) and P(hidden)

Exercise



- A box is randomly picked and then a ball is randomly picked. If the ball is found to be red, what is the probability that it came from Box1? (Hint: Box is hidden, color is measured)

Discrete Random Variables

- $P(t, w)$: Joint Probability
- $P(w) = \sum_t P(t, w)$ a.k.a. Marginalization or Sum Rule
- $P(t) = \sum_w P(t, w)$
- $P(w|t) = \frac{P(w,t)}{P(t)}$: Conditional Probability
- $P(w, t) = P(w)P(t)$: Independent Random Variables

Continuous Random Variable

- $p_X\big(x \in (a,b)\big) = \int_a^b p_X(x)dx$: pdf or density function
  - $\int p(x)dx = 1$
  - $p(x) \geq 0 \; \forall x$
- $P_X(x) = \int_{-\infty}^{x} p_X(z)dz$: CDF or distribution function
- $p(x) = \int p(x,y)dy$ Sum Rule
- $p(x,y) = p(x|y)p(y)$ Product Rule

Expectations

- Expected value of $f(x)$ Mean=μ
  - $\mathbb{E}[f(x)] = \int f(x)p(x)dx$
  - $\mathbb{E}[f(t)] = \sum_t f(t)p(t)$
- Variance of $f(x)$ Variance=σ^2
  - $\mathrm{var}[f(x)] = \mathbb{E}[\,(f(x) - \mathbb{E}[f(x)])^2\,]$ Standard Deviation=σ
  - $\mathrm{var}[f(t)] = \mathbb{E}[\,(f(t) - \mathbb{E}[f(t)])^2\,]$

Samples vs pdf

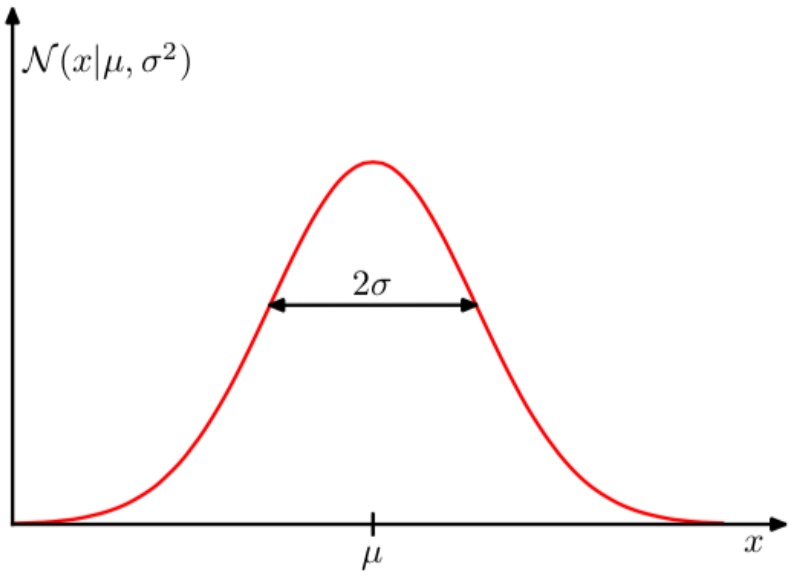
- Samples $\{s_1, s_2, s_3, \dots, s_{50}\}$ obtained by measuring x
- $E[f(x)] = \int f(x)p(x)dx$
- $E[f(x)] \approx \frac{1}{50}\sum_i f(s_i)\,; \quad s_i \sim p(x)$ (Monte Carlo approx.)
- Similarly for discrete random variable,
  - $E[f(t)] \approx \frac{1}{50}\sum_i f(t_i)\,; \quad t_i \sim p(t)$

Standard distributions

- Known or easy-to-compute properties. E.g.,
  - integration
  - marginalization
  - derivatives
- Examples
  - Bernoulli, Multinomial, etc. (discrete random variables)
  - Gaussian, Laplacian, etc. (continuous random variables)

Gaussian Distribution

$$\mathcal{N}\left(x|\mu,\sigma^2\right) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$$



$$\mathbb{E}[x] = \int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu,\sigma^2\right) x \, \mathrm{d}x = \mu.$$

$$\mathbb{E}[x^2] = \int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu,\sigma^2\right) x^2 \, \mathrm{d}x = \mu^2 + \sigma^2.$$

Multivariate Gaussian Distribution

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$$

$$\mathrm{cov}[\mathbf{x}] = \mathbb{E}\left[(\mathbf{x}-\mathbb{E}[\mathbf{x}])(\mathbf{x}-\mathbb{E}[\mathbf{x}])^{\mathrm{T}}\right]$$

$$\mathrm{cov}[\mathbf{x}] = \boldsymbol{\Sigma}$$

$$\mathbb{E}[\mathbf{x}\mathbf{x}^{\mathrm{T}}] = \boldsymbol{\mu}\boldsymbol{\mu}^{\mathrm{T}} + \boldsymbol{\Sigma}$$

References

Pattern Recognition and Machine Learning, Bishop, Springer (Available online: http://research.microsoft.com/~cmbishop/PRML)

- Section 1.2
- Chapter 2